Using Proposition 6.3, we know that for a Wiener process $\mathrm{W}(t)$, $\mathrm{Y}(t) \stackrel{\text{def}}{=} \int_0^t \exp(\gamma s)\, \mathrm{dW}(s)$ is a Gaussian random variable with mean zero and variance

$$\mathrm{Var}\{\mathrm{Y}(t)\} = \int_0^t e^{2\gamma s} \mathrm{d}s = \frac{e^{2\gamma t} - 1}{2\gamma}.$$

The solution with equivalent distribution is therefore

$$\mathrm{X}(t) = e^{-\gamma t}\mathrm{X}(0) + \sigma\sqrt{(2\gamma)^{-1}(1 - e^{-2\gamma t})}\hat{\mathrm{Y}}(t).$$

where $\hat{\mathrm{Y}}(t)$ should be interpreted as random process which at every time $t$ evaluates to a normal random variable ($\hat{\mathrm{Y}}(t) \sim N(0, 1)$). This is what we mean by the exact distributional solution of the Ornstein-Uhlenbeck SDE with prescribed initial value $\mathrm{X}(0)$.

Let $\sigma = \sqrt{2\gamma \mathrm{k_B T} m}$, then for large $t$ the exponentials of the form $e^{-\gamma t}$ tend to zero, and we have

$$\mathrm{X}(t) \approx \sqrt{k_\mathrm{B} \mathrm{T} m}\hat{\mathrm{Y}}(t),$$

in which case $\mathrm{X}(t)$ is drawn from the probability density

$$\rho_\infty(u) = \frac{1}{\sqrt{2\pi m \mathrm{k_B T}}} \exp\left(-\frac{1}{\mathrm{k_B T}} \frac{u^2}{2m}\right).$$

If we think of $\mathrm{X}(t)$ as not the position, but actually the momentum, then the Ornstein-Uhlenbeck equation samples, in the long term, the Gibbs-Boltzmann distribution associated to a single particle in a thermal bath at temperature T. In recognition of this, from this point forward, we will replace X with $p$ as the variable in the Ornstein-Uhlenbeck equation.[7] For a system with $\mathrm{N_c}$ configurational variables we may write

$$\mathrm{d}p_i = -\gamma p_i \mathrm{d}t + \sqrt{2\gamma \mathrm{k_B T} m_i}\mathrm{dW}_i, \tag{6.20}$$

where $\mathrm{W}_1(t), \mathrm{W}_2(t), \ldots, \mathrm{W}_{\mathrm{N_c}}(t)$ represent $\mathrm{N_c}$ independent Wiener processes, or, in vectorial notation, denoting $\mathbf{W} = [\mathrm{W}_1, \mathrm{W}_2, \ldots, \mathrm{W}_{\mathrm{N_c}}]^T$, we have

$$\mathrm{d}\boldsymbol{p} = -\gamma \boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma \mathrm{k_B T}}\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}. \tag{6.21}$$

[7] In so doing, we must in many cases give up the traditional notational convention of indicating solutions of SDEs by capital letters and the corresponding variables in the density function by the lower case equivalent. This however allows us to maintain better notational coherence with the earlier treatment of deterministic systems.